Project Report

Phone Addiction Predictor Using Machine Learning

Introduction

Smartphones have become an important part of daily life, especially for students. They are used for communication, education, entertainment, and social media. However, excessive phone usage can lead to serious problems such as lack of sleep, reduced productivity, poor concentration, and mental stress. One of the growing concerns today is phone addiction, where a person becomes overly dependent on using their mobile device.

This project, Phone Addiction Predictor, was created to solve this problem using Machine Learning. The aim of the project is to predict whether a person is addicted to their phone or not based on a few daily usage habits. The project uses a simple machine learning model to analyze user behavior and classify the result as Addicted (1) or Not Addicted (0). The implementation and project description are based on the uploaded code and documentation

Problem Chosen and Why It Matters

The problem chosen for this project is phone addiction. This issue is highly relevant because many students and young people spend several hours daily on their phones without realizing how much it affects their health and routine.

Phone addiction matters because it can lead to:

- Poor sleep habits
- Lack of focus on studies or work
- Increased distraction
- Reduced face-to-face interaction
- Mental stress and dependency

As smartphones become more common, it is important to identify unhealthy usage patterns early. This project tries to address that by using data and machine learning to make a simple prediction.

Approach to Solving the Problem

To solve this problem, I created a machine learning-based predictor that uses user behavior data to determine addiction status. The model uses three important features:

1. **Screen Time** – How many hours a person spends on their phone
2. **Sleep Hours** – How many hours a person sleeps
3. **App Usage** – How frequently or heavily apps are used

These factors were chosen because they are strongly related to mobile phone dependency. A person with high screen time, low sleep hours, and heavy app usage is more likely to show signs of addiction.

The target column in the dataset is:

- **Addicted →**
    - **1 = Addicted**
    - **0 = Not Addicted**

The project uses Python along with the libraries Pandas and Scikit-learn to load the dataset, train the model, test its performance, and make predictions.

How the Project Works

The project follows a simple machine learning workflow.

1. Loading the Dataset

The dataset is read from a CSV file using Pandas. It contains the values of screen time, sleep hours, app usage, and addiction label.

2. Selecting Features and Target

The input values used for prediction are:

- ScreenTime
- SleepHours
- AppUsage

The output is:

- Addicted

3. Splitting the Data

The dataset is divided into:

- 80% training data

- 20% testing data

This helps the model learn from one part of the data and then get tested on another part.

4. Training the Model

The project uses Logistic Regression, which is a simple and suitable machine learning algorithm for classification problems.

5. Testing Accuracy

After training, the model checks how accurately it predicts addiction on test data.

6. Making a Prediction

The code also includes a sample prediction using values such as:

- Screen Time = 9
- Sleep Hours = 5
- App Usage = 20

The model then predicts whether the person is addicted or not. This complete workflow is implemented in the uploaded Python file

Key Decisions Made

Several important decisions were made while building this project.

Choosing a Relevant Topic

The first decision was selecting **phone addiction** as the problem because it is a real and growing issue in student life.

Keeping the Project Simple

Instead of making the project too complex, I decided to keep it beginner-friendly and understandable. This made the project easier to build, explain, and present.

Using Only Three Features

I selected only three input factors:

- Screen Time
- Sleep Hours
- App Usage

These were chosen because they are easy to understand and directly connected to phone usage behavior.

Choosing Logistic Regression

Logistic Regression was selected because:

- It works well for binary classification
- It is simple to implement
- It is suitable for beginner machine learning projects

This was an effective decision for a project of this level.

Challenges Faced

While working on this project, I faced a few challenges.

1. Choosing the Right Features

Phone addiction can depend on many things such as social media use, gaming habits, emotional dependency, and study interruptions. Choosing only three features made the project simple, but it was also a limitation.

2. Understanding the Dataset

A machine learning model depends heavily on the dataset. It was important to understand which columns should be used as input and which should be used as the output.

3. Model Accuracy Depends on Data

The performance of the model depends on how good and balanced the dataset is. If the data is limited or unrealistic, the prediction may not be fully reliable.

4. No User Interface

The project currently runs in code form only. A non-technical user cannot directly use it unless a proper interface is added.

Even with these challenges, the project was successfully completed and gave meaningful results.

What I Learned

1. Basics of Machine Learning

I learned how a machine learning project is built step by step:

- Data loading
- Feature selection
- Data splitting

- Model training
- Accuracy checking
- Prediction

2. Real-World Use of AI

I understood how AI and machine learning can be used to solve everyday problems, not just technical or industrial ones.

3. Importance of Data

I learned that the quality of a machine learning project depends greatly on the quality of the dataset and the selected features.

4. Simplicity Can Still Be Effective

This project showed me that even a simple model like Logistic Regression can be useful when applied to a meaningful problem.

5. Documentation and Presentation

I also learned that a good project is not only about writing code but also about explaining the idea, structure, and outcome clearly. The README file helped organize the project properly

Conclusion

The Phone Addiction Predictor is a simple but meaningful machine learning project that addresses an important modern problem. It predicts whether a person is likely to be addicted to their phone based on screen time, sleep hours, and app usage.

This project is useful because it combines:

- A real-life issue
- Data analysis
- Machine learning
- Prediction

Although the project has some limitations, it provides a strong foundation for future improvements such as adding more features, better evaluation methods, and a graphical user interface.

Overall, this project helped me understand how machine learning can be used in practical and socially relevant ways.